

PySpark Installation Guide for Mac OS

Follow the steps below to install Spark on Mac OS and configure Python for PySpark.

Installing Prerequisites:

1. Java

PySpark requires Java 1.8.x

Check if Java is already available by opening the command prompt and giving the command as shown here.

```
$ java -version
java version "1.8.0_471"
Java(TM) SE Runtime Environment (build 1.8.0_471-b09)
Java HotSpot(TM) 64-Bit Server VM (build 25.471-b09, mixed mode)
```

If not available, then download the .DMG file of the JDK for Mac OS from official Oracle website from the link below and install it by double-clicking to run the downloaded file.

<https://www.oracle.com/in/java/technologies/javase/javase8u211-later-archive-downloads.html>

After installing check as shown above and make sure the correct version is installed and available. Also, make a note of the folder where it is installed as we will need the path later.

2. Python

Python 3.8.x to 3.11.x is required to run PySpark. Note that Python versions above 3.11.x are not compatible with current version of PySpark as of now.

Check if a compatible version of Python is already available, from the command prompt as shown here.

```
$ python --version
Python 3.10.10
```

If not available and if Anaconda is the preferred environment then you can download it from the link below and install it.

<https://repo.anaconda.com/archive/Anaconda3-2025.12-2-MacOSX-arm64.pkg>

If Anaconda is already installed then make sure that it is added to the path.

Installing Spark:

Download Spark from Apache Spark web site:

<https://spark.apache.org/downloads.html>

In the dropdown *Choose a Spark release*: select a release 3.5.X

In the dropdown *Choose package type*: Pre-built for Apache Hadoop 3.3 and later

Download Apache Spark™

1. Choose a Spark release: ▼
2. Choose a package type: ▼
3. Download Spark: [spark-3.5.8-bin-hadoop3.tgz](#)
4. Verify this release using the 3.5.8 [signatures](#), [checksums](#) and [project release KEYS](#) by following these [procedures](#).

You can use the file from the Downloads directory.

Optionally you can copy the above file into a system-wide common directory like `/usr/local/`
Make sure not to use spaces in the folder names.

Or you can use a directory within your home director such as `/home/<your login>/`

You can now extract the files from the archive using the command below.

```
$ tar -zxvf spark-3.5.x-bin-hadoop3.tgz
```

Rename the directory.

```
$ mv spark-3.5.x-bin-hadoop3 spark
```

Make a note of the directory name where it is installed as we will need the path later.

Setting up Environment Variables:

Environment variables need to be set so that PySpark can look up the correct directories for the required files.

Add each of the new variables listed below.

1. `export JAVA_HOME=<full path of Java installation directory>`
(Only if not present already)
2. `export SPARK_HOME=<full path of the above directory where spark files are extracted>`
3. `export PYSARK_PYTHON=python`

4. `export PATH=$PATH:$SPARK_HOME/bin`
5. `export PATH=$PATH:$JAVA_HOME/bin`
(Only if not present already)

This completes the installation of Spark on Mac OS.

You can run PySpark shell from the command prompt by giving the command `pyspark` and verify.